ROUTING USING

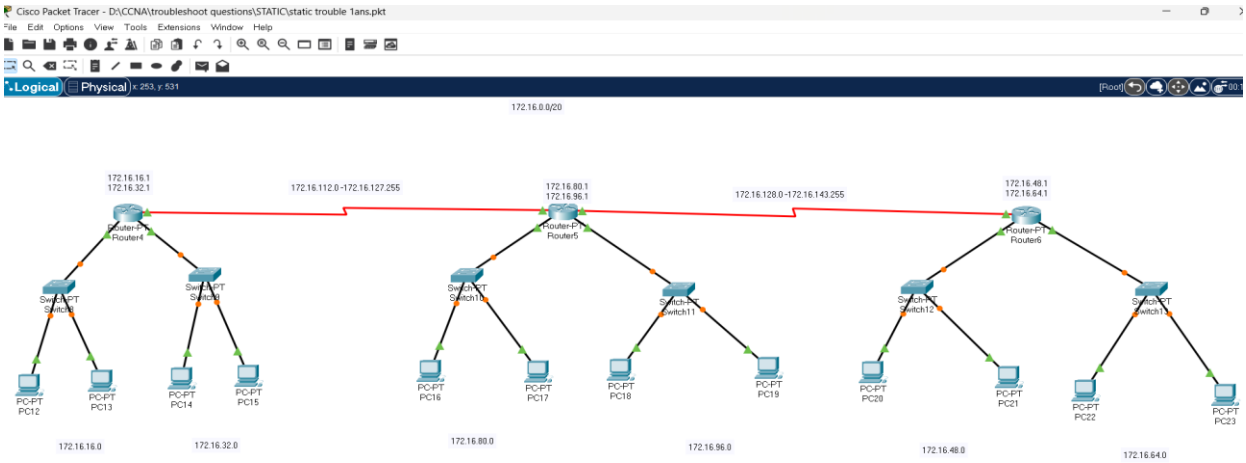
CISCO PACKET TRACER

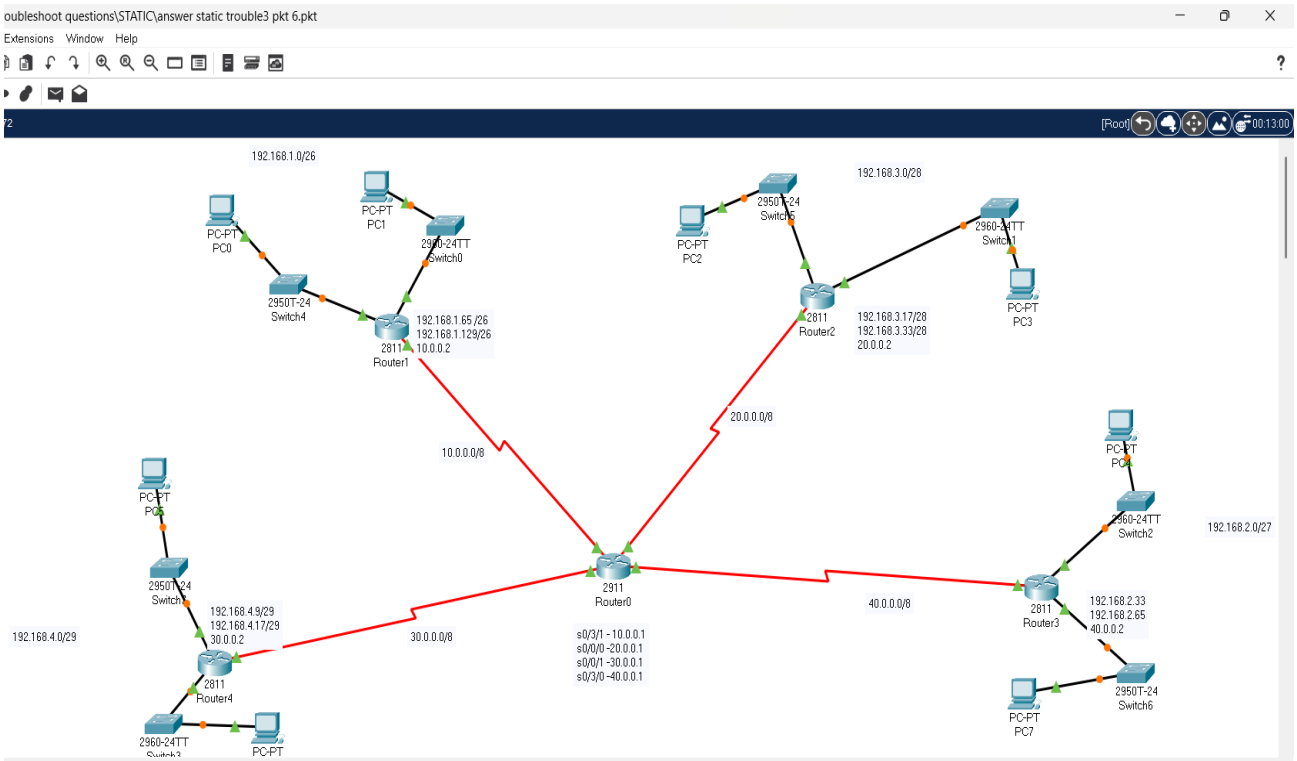
This project focuses on designing and simulating a computer network using Cisco Packet Tracer to demonstrate the implementation and comparison of various routing techniques. The network topology is built to support multiple routing protocols, including static routing, RIP (Routing Information Protocol), and EIGRP (Enhanced Interior Gateway Routing Protocol).

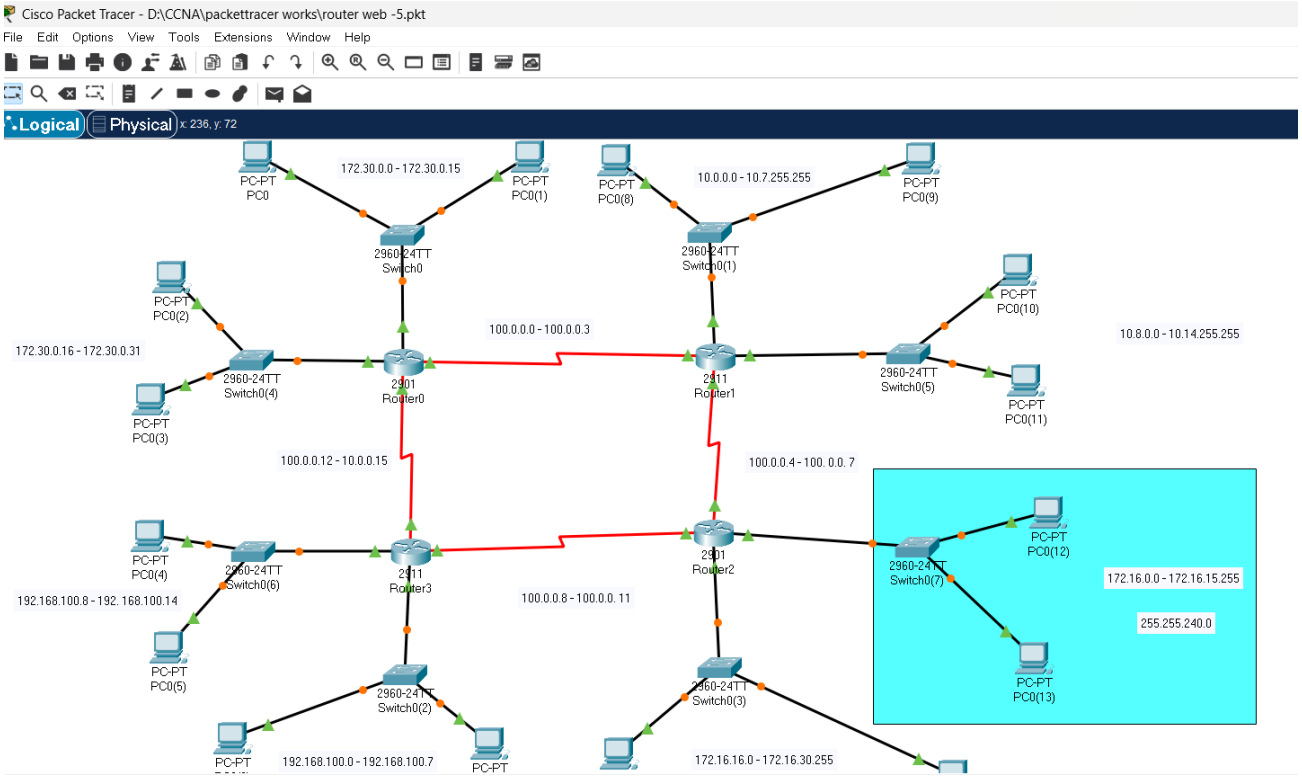
The objective of the project is to understand how different routing methods operate, configure them in a simulated environment, and analyze their performance in terms of convergence time, scalability, and efficiency. Static routing is implemented to establish manual path selection, while dynamic routing protocols such as RIP and EIGRP are configured to enable automatic route discovery and adaptability.

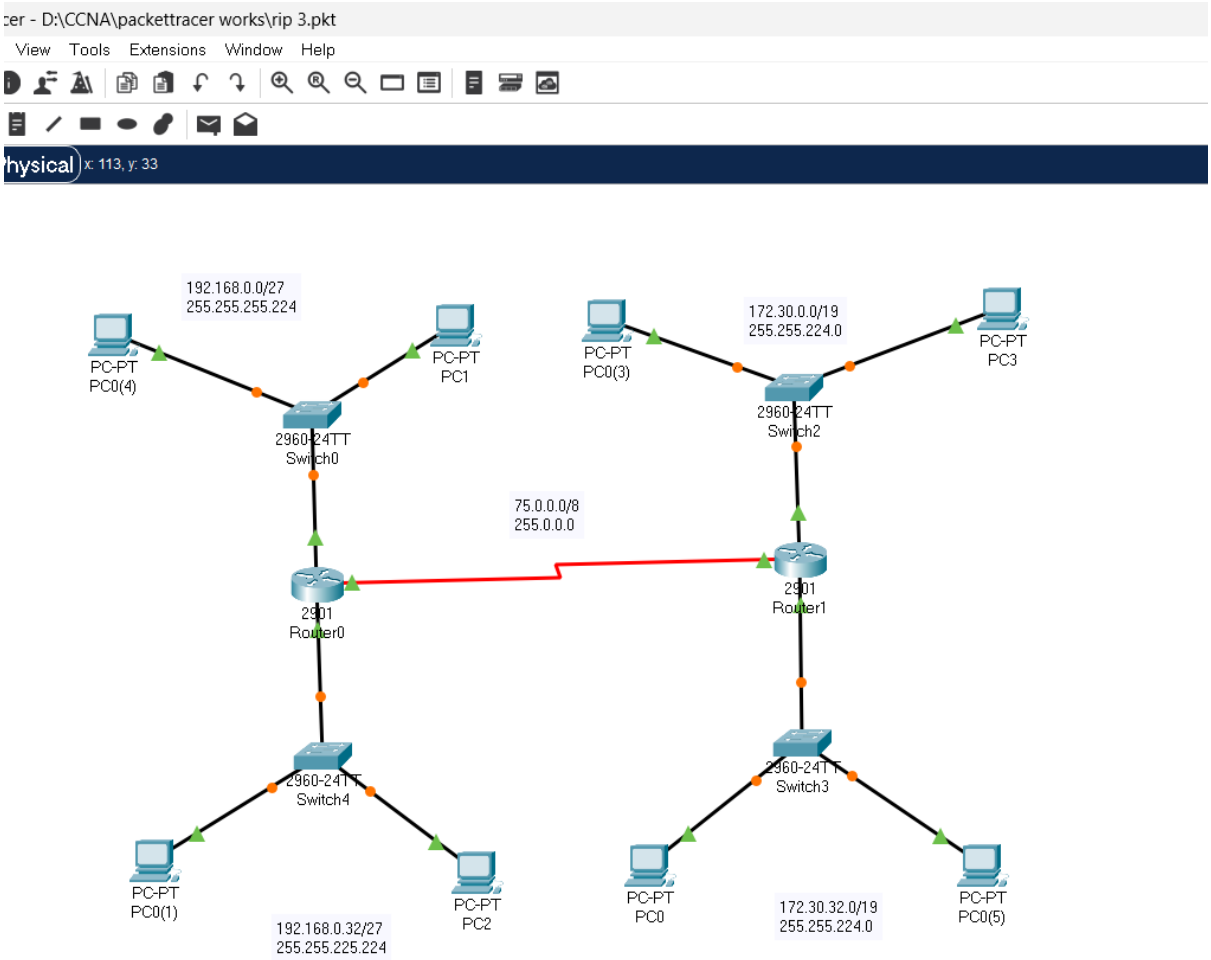
Through this project, users gain hands-on experience in network configuration, troubleshooting, and protocol behavior. It also helps in identifying the advantages and limitations of each routing technique in real-world networking scenarios.

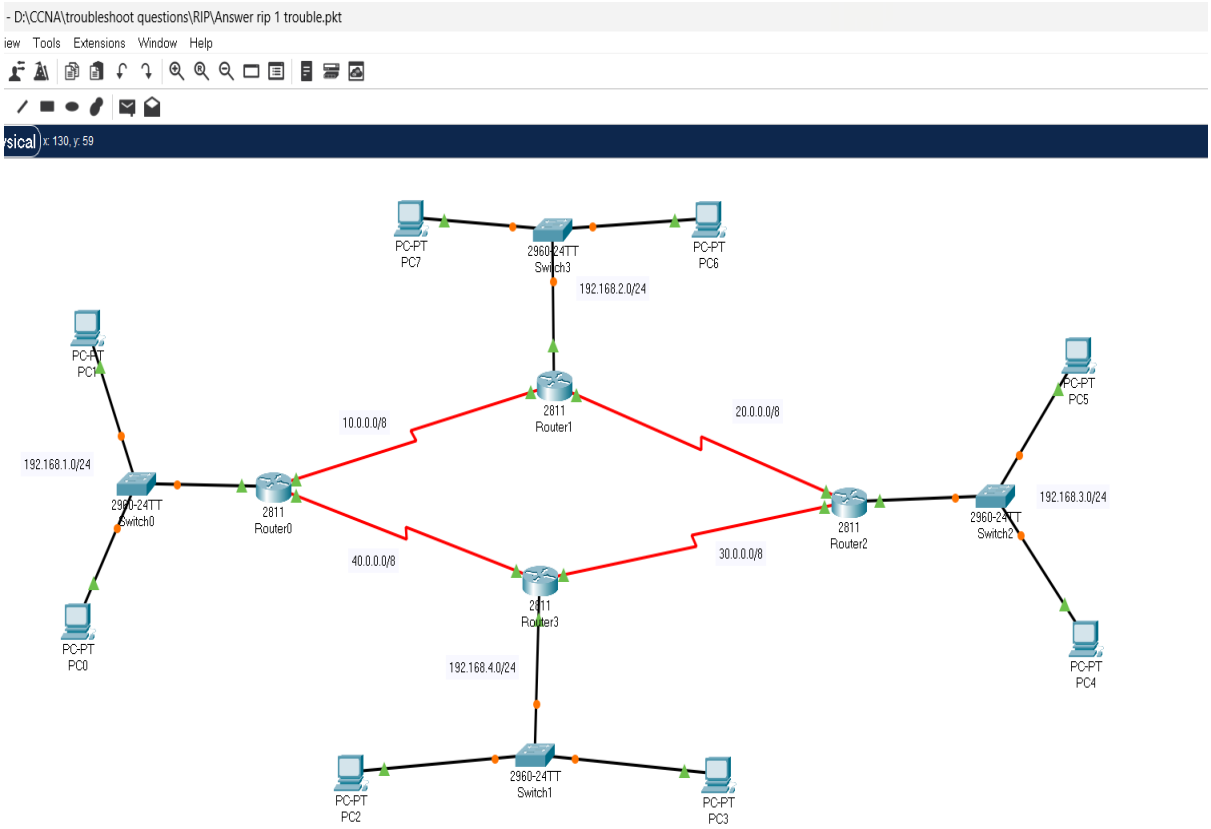
Screenshots provided below

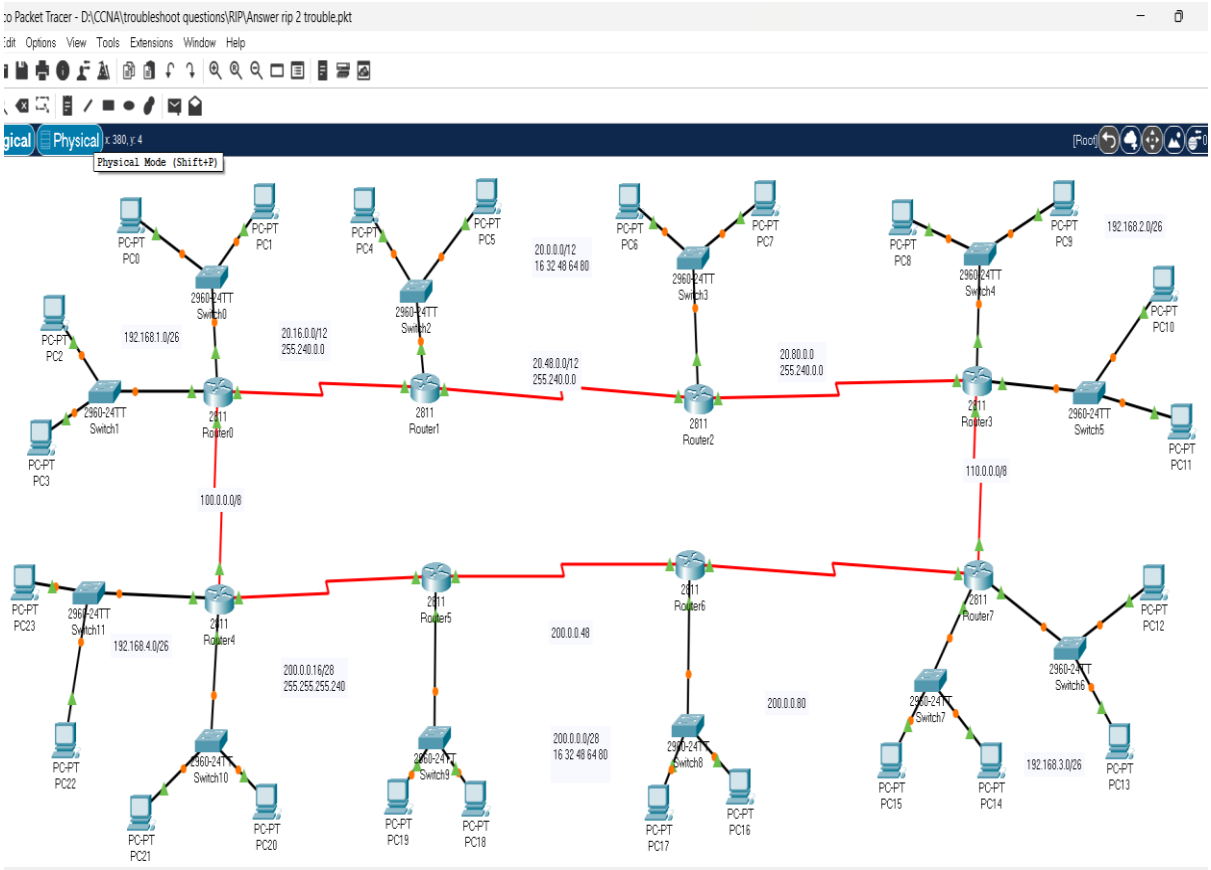
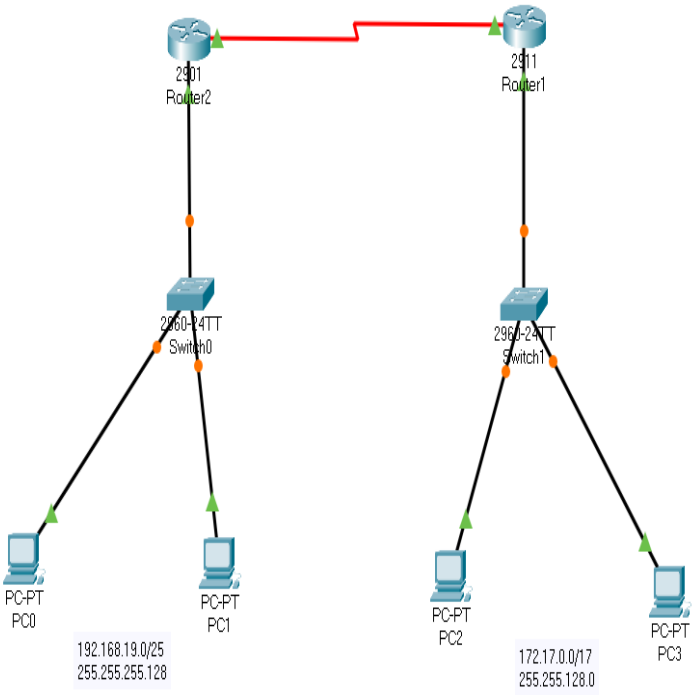



oubleshoot questions\STATIC\answer static trouble3 pkt 6.pkt
Extensions Window Help
[Root]
00:13:00
192.168.1.0/26
PC-PT PC0
PC-PT PC1
2960-24TT Switch0
2950T-24 Switch4
192.168.1.65/26
192.168.1.129/26
10.0.0.2
2811 Router1
192.168.3.0/28
PC-PT PC2
2950T-24 Switch5
2960-24TT Switch1
PC-PT PC3
2811 Router2
192.168.3.17/28
192.168.3.33/28
20.0.0.2
10.0.0.0/8
20.0.0.0/8
PC-PT PC4
2960-24TT Switch2
192.168.2.0/27
2911 Router0
40.0.0.0/8
2811 Router3
192.168.2.33
192.168.2.65
40.0.0.2
2950T-24 Switch6
PC-PT PC7
PC-PT PC5
2950T-24 Switch5
192.168.4.9/29
192.168.4.17/29
30.0.0.2
192.168.4.0/29
30.0.0.0/8
s0/3/1 - 10.0.0.1
s0/0/0 -20.0.0.1
s0/0/1 -30.0.0.1
s0/3/0 -40.0.0.1
2811 Router4
2960-24TT Switch3
PC-PT


Cisco Packet Tracer - D:\CCNA\packettracer works\router web -5.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 236, y: 72
PC-PT PC0
172.30.0.0 - 172.30.0.15
PC-PT PC0(1)
PC-PT PC0(8)
10.0.0.0 - 10.7.255.255
PC-PT PC0(9)
2960-24TT Switch0
2960-24TT Switch0(1)
PC-PT PC0(2)
PC-PT PC0(10)
172.30.0.16 - 172.30.0.31
2960-24TT Switch0(4)
2901 Router0
100.0.0.0 - 100.0.0.3
2911 Router1
2960-24TT Switch0(5)
10.8.0.0 - 10.14.255.255
PC-PT PC0(3)
PC-PT PC0(11)
100.0.0.12 - 10.0.0.15
100.0.0.4 - 100. 0.0. 7
PC-PT PC0(4)
2960-24TT Switch0(6)
2911 Router3
2901 Router2
2960-24TT Switch0(7)
PC-PT PC0(12)
192.168.100.8 - 192. 168.100.14
100.0.0.8 - 100.0.0. 11
172.16.0.0 - 172.16.15.255
255.255.240.0
PC-PT PC0(5)
PC-PT PC0(13)
2960-24TT Switch0(2)
2960-24TT Switch0(3)
PC-PT
192.168.100.0 - 192.168.100.7
PC-PT
172.16.16.0 - 172.16.30.255


cer - D:\CCNA\packettracer works\rip 3.pkt
View Tools Extensions Window Help
hysical x: 113, y: 33
192.168.0.0/27
255.255.255.224
PC-PT
PC0(4)
PC-PT
PC1
2960-24TT
Switch0
PC-PT
PC0(3)
172.30.0.0/19
255.255.224.0
PC-PT
PC3
2960-24TT
Switch2
75.0.0.0/8
255.0.0.0
2901
Router0
2901
Router1
2960-24TT
Switch4
2960-24TT
Switch3
PC-PT
PC0(1)
PC-PT
PC2
192.168.0.32/27
255.255.225.224
PC-PT
PC0
172.30.32.0/19
255.255.224.0
PC-PT
PC0(5)


- D:\CCNA\troubleshoot questions\RIP\Answer rip 1 trouble.pkt
iew Tools Extensions Window Help
sical x: 130, y: 59
PC-PT
PC7
2960-24TT
Switch3
PC-PT
PC6
192.168.2.0/24
PC-PT
PC1
2811
Router1
PC-PT
PC5
10.0.0.0/8
20.0.0.0/8
192.168.1.0/24
2960-24TT
Switch0
2811
Router0
2811
Router2
2960-24TT
Switch2
192.168.3.0/24
40.0.0.0/8
30.0.0.0/8
2811
Router3
PC-PT
PC0
PC-PT
PC4
192.168.4.0/24
2960-24TT
Switch1
PC-PT
PC2
PC-PT
PC3



110.0.0.0/9
255.128.0.0